

RETAILPULSE

Business Problem Statement & Analytical Scope

Real-World Retail Transaction Analytics Project

1. Executive Context

Retail organizations generate large volumes of transaction data containing sales, products, customers, dates, quantities, prices and geographic information. RetailPulse transforms imperfect transaction data into decision-ready intelligence covering sales performance, customer retention, demand planning and inventory decisions.

2. Business Problem

The business needs a reliable analytical solution to understand revenue and order performance, customer value and inactivity risk, product demand, future demand expectations, and inventory replenishment signals. The challenge is to move from a damaged and inconsistent raw transaction file to trustworthy metrics and actionable Power BI reporting.

3. Data Quality Findings

Metric	Observed value
Total records	1,067,371
Total columns	9
Duplicate records	34,335
Missing CustomerID	243,007
Missing Description	4,382
Unique customers	5,942
Unique invoices	53,628
Unique products	5,305
Unique countries	43
Cancellation records	19,494
Negative quantity records	22,950
Zero quantity records	0
Zero price records	6,202
Negative price records	5
Invalid dates	0

4. Business Questions

- How is revenue and order activity changing over time?
- Which countries and products contribute the most revenue and quantity?
- Which customer segments generate the greatest value?
- Which customers show signs of inactivity or churn risk?
- Which products show meaningful demand patterns?
- What demand is expected over the next 30 days?
- How can forecasted demand support safety-stock and reorder-point planning?

5. Analytical Scope

Workstream	Purpose	Output
Data Profiling	Assess completeness, duplicates and anomalies	Data quality profile
Data Cleaning	Create analysis-ready data	Cleaned datasets
EDA	Identify sales, customer, product and geographic patterns	Business analysis
RFM Segmentation	Measure customer value	Customer segments
Churn / Inactivity	Identify retention priorities	Risk analysis
Demand Forecasting	Estimate short-term demand	30-day forecast
Inventory Optimization	Translate demand into planning signals	Safety stock / reorder guidance
Power BI	Communicate decisions	Interactive dashboard

6. Data Treatment Principles

- Cancellations and negative quantities must be interpreted in business context rather than blindly deleted.
- Missing CustomerID values should be retained where useful for product/revenue analysis, while customer-level analysis should use identifiable customers.
- Missing descriptions should not be invented; product identifiers can support analysis where descriptions are unavailable.
- Duplicates, zero prices and negative prices should be validated before KPI reporting.
- Forecast and inventory outputs are planning guidance unless verified warehouse and supplier data are available.

7. Expected Business Value

RetailPulse is designed to provide a single analytical view of commercial performance and customer behavior while connecting demand forecasting to inventory planning. The intended value is faster identification of revenue drivers, clearer prioritization of high-value customers, better demand visibility and more structured replenishment decisions.

8. Technology Stack

Python: Pandas, NumPy and analytical/forecasting libraries. **Jupyter:** profiling, cleaning, EDA, segmentation, churn, forecasting and inventory analysis. **Power BI:** Power Query, DAX, interactive visuals and executive reporting.